Current-Starved Ring Oscillator

Silicon Jackets AMS Onboarding

Ash Sridhar

Design Targets

Specification window the design must satisfy at the tt corner

5	500 MHz	<500 µW	1.8 V
Ring stages (N)	Target f_osc (450–550 MHz window)	Oscillator power budget	VDD

Process & device parameters (given)

Parameter	NFET	PFET
Vth0 (V)	0.52	−1.06
γ (V^0.5)	0.54	0.43
uCox (µA/V²)	264	87
CGSO/CGDO (fF/µm)	0.254	0.0525
CJ (fF/µm²)	1.35	0.738
CJSWG (fF/µm)	0.233	0.239
CJSW (fF/µm)	0.036	0.0989
Default W / L (µm)	0.42 / 0.15	0.55 / 0.15

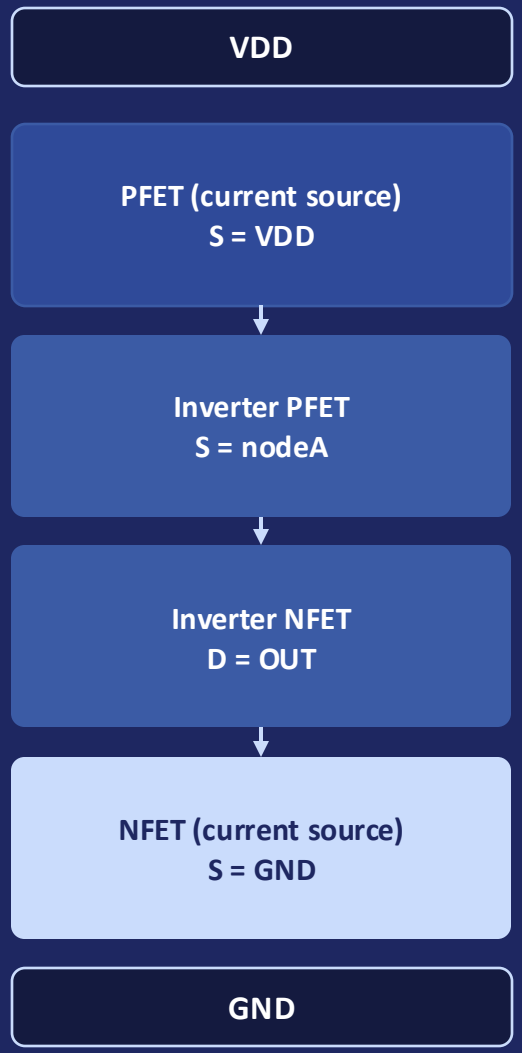
Selected design values

N (stages)	**5**
Target f_osc	**500 MHz**
VIN (selected)	**0.6 V**

Common constants

Cox	8.3 fF/µm² (both)
Ldiff	0.29 µm (both)
2φF	0.85 V (both)

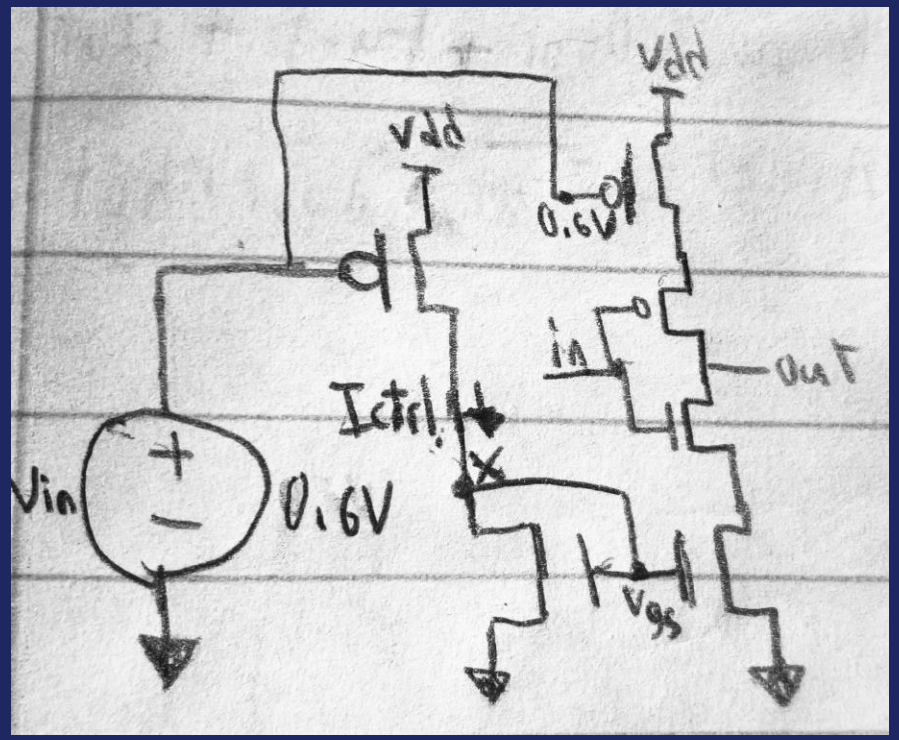
Per-Stage Topology



Four Transistors in Series

- PFET and NFET current sources above and below a standard inverter limiting how much current can flow through it.
- Holding both in saturation makes them behave like current sources; the current they pass directly sets stage delay (td = CL·VDD / 2·I_ctrl).
- N = 5 stages: an odd number of inversions around the loop prevents the ring from settling into a stable DC state.
- Because the current is set by a voltage (VIN, via a bias network), this is a true voltage-controlled oscillator (VCO).

Voltage-Controlled Bias via Current Mirror



Routing Description

1) VIN's positive terminal wires directly to a node shared by the gate of the mirror PFET and the gate of every ring-stage PFET. VIN's negative terminal goes to GND. This fixes that gate node at 0.6V, so every PFET's V_SG = VDD − 0.6V = 1.2V.

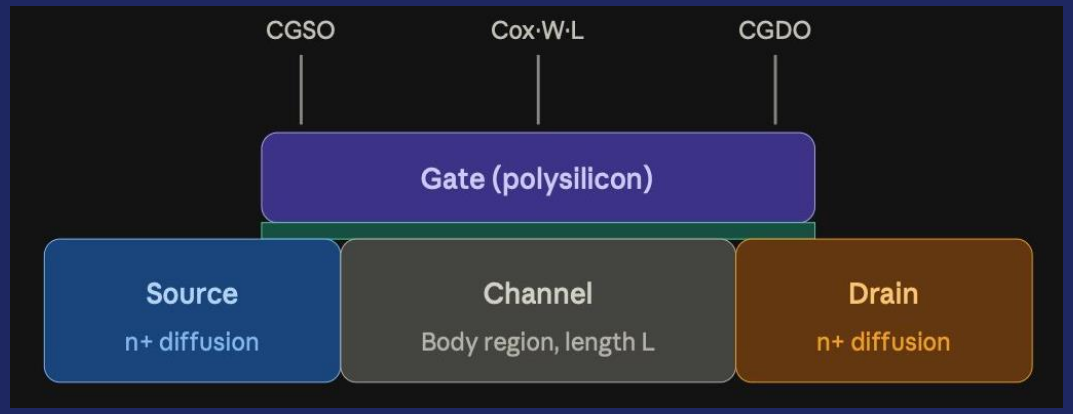
2) The mirror PFET's drain and the mirror NFET's drain and gate meet at node X. Since this is a single series path from VDD to GND with no other branches, KCL forces the mirror PFET and mirror NFET to carry the exact same current.

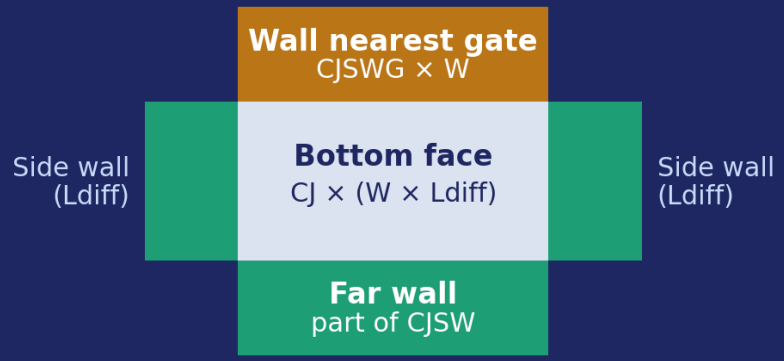
3) The mirror NFET is diode-connected which means that its gate is tied to its own drain, which makes it self-bias to whatever gate voltage sustains that same current, given its own W/L. This means that for the NFETs, we can set W/L as long as it results in a reasonable voltage at node X.

4) Every ring-stage NFET's gate is also connected to that node X, which mirrors that current into each stage's own NFET and allows VIN to result in a controlled current in every stage.

Load Capacitance & Target Current

How C_L was derived: Every OUT node is loaded by two physical contributions per transistor: the gate capacitance, which is the Cox*W*L term plus CGSO/CGDO overlap capacitance (where the source/drain diffusion extends slightly under the gate edge), and the drain junction capacitance(CJ*W* Ldiff plus CJSWG/CJSW perimeter terms split between the one edge bordering the gate and the other three edges). Summing all four terms gives the total capacitance.





Amber = CJSWG (the one edge touching the gate)
Teal = CJSW group: far wall + 2 side walls

2.138 fF
Total C_L

9.621 µA
Target I_ctrl

```
C_gate,n = Cox·Wn·Ln + (CGSO_n+CGDO_n)·Wn
         = 8.3(0.42)(0.15) + 0.508(0.42)
         = 0.523 + 0.213 = 0.736 fF

C_gate,p = Cox·Wp·Lp + (CGSO_p+CGDO_p)·Wp
         = 8.3(0.55)(0.15) + 0.105(0.55)
         = 0.685 + 0.058 = 0.743 fF

C_drain,n = CJ_n(Wn·Ldiff) + CJSWG_n·Wn
           + CJSW_n(2·Ldiff+Wn)
          = 1.35(0.122)+0.233(0.42)+0.036(1.00)
          = 0.164+0.098+0.036 = 0.298 fF

C_drain,p = CJ_p(Wp·Ldiff) + CJSWG_p·Wp
           + CJSW_p(2·Ldiff+Wp)
          = 0.738(0.160)+0.239(0.55)+0.0989(1.13)
          = 0.118+0.131+0.112 = 0.361 fF

C_L = 0.736+0.743+0.298+0.361 = 2.138 fF

I_ctrl = f_osc·N·C_L·VDD
  = (500e6)(5)(2.138e-15)(1.8) = 9.621 µA
```

Solving for W/L: PFET & NFET Equation

PFET equation

$$V_{SG} = VDD - VIN$$
$$W/L = 2 \cdot I_{ctrl} / [uCox_p \cdot (V_{SG} - |Vth,p|)^2]$$

With VIN = 0.6V:

$V_{SG} = 1.8 - 0.6 = 1.2\ V$

$W/L = 2(9.621) / [87 \cdot (1.2 - 1.06)^2]$

$W/L = 19.242 / 1.7052$

$W/L \approx 11.284$

No Body Effect

The W/L we are calculating is for the top-most and bottom most PFETs and NFETS that act as the current sources, as well as the PFET and NFET in the current mirror. All PFETS have both source and body connected to VDD, so the Vth,p is equal to the Vth0,p. All NFETS have both source and body connected to GND, so the Vth,n is equal to the Vth0,n.

NFET W/L Choice

The NFET's gate voltage depends on node X, which can be set by changing the W/L of the NFET. Therefore, we can leave the W/L of the NFET as default(2.8) and get the correct I_ctrl value since the node X automatically adjusts to the right voltage.

NFET equation

$$W/L = 2 \cdot I_{ctrl} / [uCox_n \cdot (V_{GS,n} - Vth,n)^2]$$
$$\Rightarrow V_{GS,n} - Vth,n = \sqrt{(2 \cdot I_{ctrl} / (uCox_n \cdot (W/L)))}$$
$$V_{GS,n} - Vth,n = \sqrt{(19.242 / (264 \cdot 2.8))}$$
$$V_{GS,n} - Vth,n \approx 0.161\ V$$

Gate voltage(node X) needed:

$V_{GS,n} = Vth,n + 0.161$

$V_{GS,n} = 0.52 + 0.161 \approx$ **$0.681\ V$**

Summary

C_L	**2.138 fF**
I_ctrl	**9.621 µA**
VIN	**0.6 V**
(W/L) PFET / mirror PFET	**11.284**
(W/L) NFET / mirror NFET	**2.8**
PFET : NFET size ratio	**≈ 4.02: 1**

Stage 2

Revisions from Simulation

Original Submission 1 calculations unchanged — sizing revised after schematic-level simulation

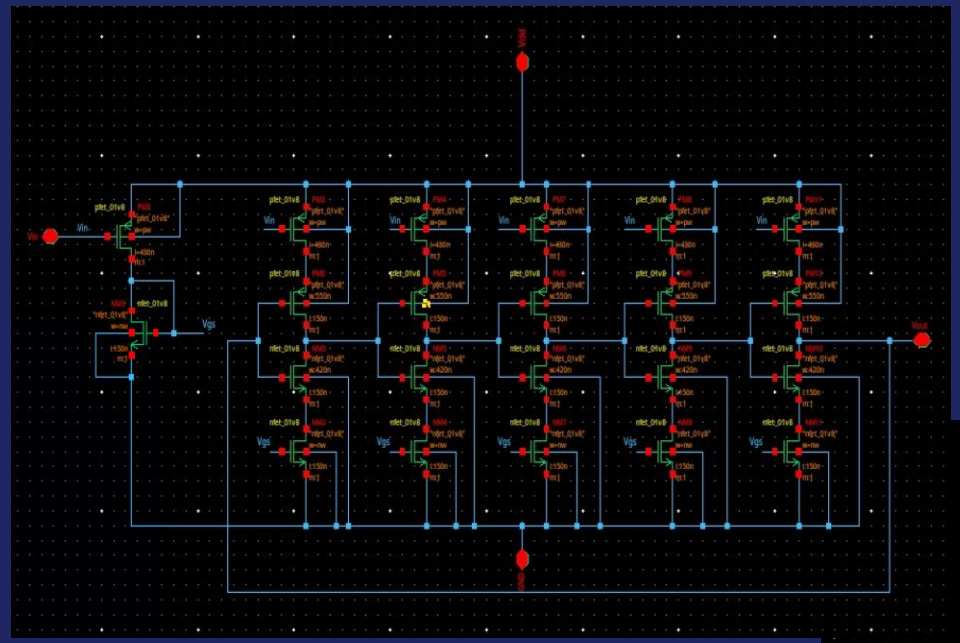
Simulation showed the real Sky130 PMOS threshold shifts by roughly 160mV across the 0–70°C spec range and because the current-source PFET's overdrive is set directly by VIN and this threshold that 160mV shift controlled the bias point and pushed the oscillation frequency outside the 450–550MHz window at temperature extremes.

The fix was to lower VIN and reduce the PFET/mirror sizing together, which raises the transistors' operating overdrive while holding the target current — and therefore frequency — fixed at 27°C. A larger overdrive baseline makes the same fixed 160mV threshold shift a much smaller part of the bias point, which is what stabilizes frequency across temperature.

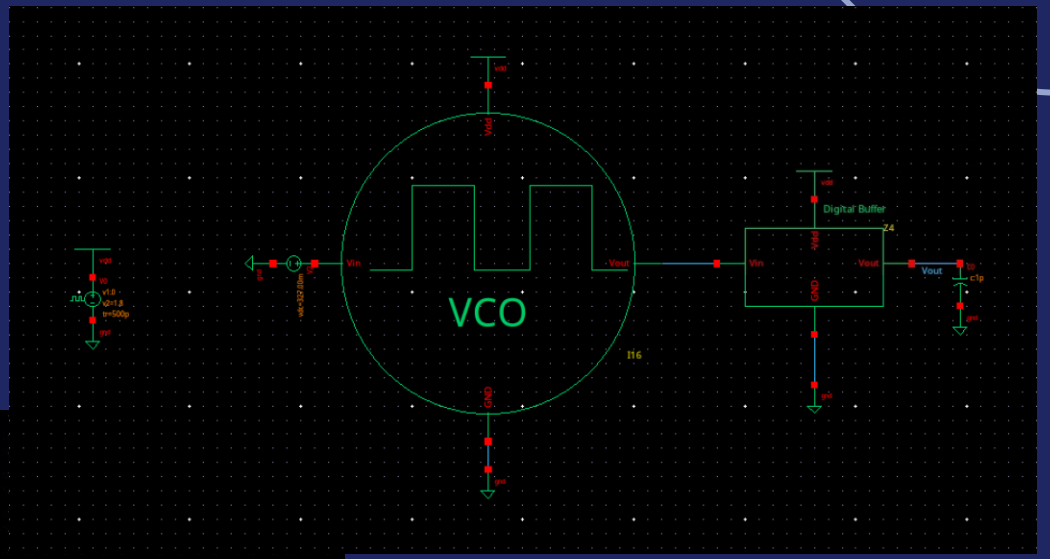
Final values: PFETs: W=420nm, L=430nm, NFETs are unchanged, with VIN=0.327V. This meets 450–550MHz across the full 0–70 C range.

Design Schematics

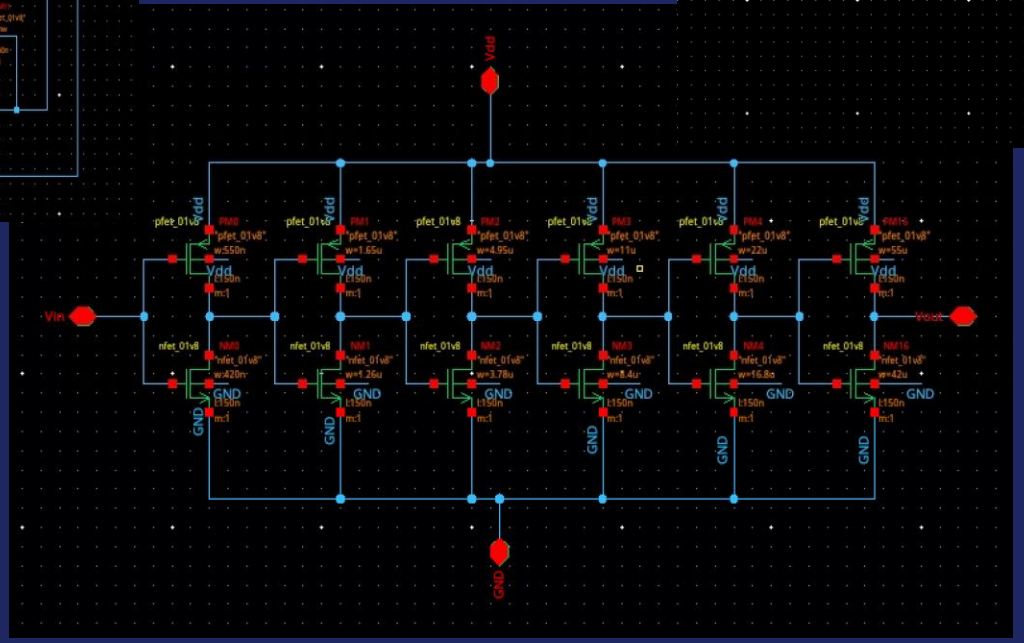
Ring oscillator, output buffer, and testbench — Cadence Virtuoso



Ring Oscillator (5 stages)



Testbench (sources + load)



Output Buffer (6-stage taper)

Spec Screenshots

	Expression	Value 0 C	Value 27 C	Value 70 C
Freq	0 average(freq(clip(VT("/Vout") 30n 50n) "rising" ?xName "time" ?mode "auto" ?threshold 0.9))	451.2E6	490.0E6	549.6E6
Rise Time	average(riseTime(clip(VT("/Vout") 30n 50n) 0 nil 1.8 nil 10 90 t "time"))	205.6E-12	205.7E-12	202.2E-12
Fall Time	0 average(fallTime(clip(VT("/Vout") 30n 50n) 1.8 nil 0 nil 10 90 nil "time"))	104.9E-12	110.7E-12	125.3E-12
Duty Cycle	0 average(dutyCycle(clip(VT("/Vout") 30n 50n) ?...	51.66	51.19	51.01
Power	) average(clip(IT("/V5/PLUS") 30n 50n))*1.8	37.63E-6	41.41E-6	47.53E-6

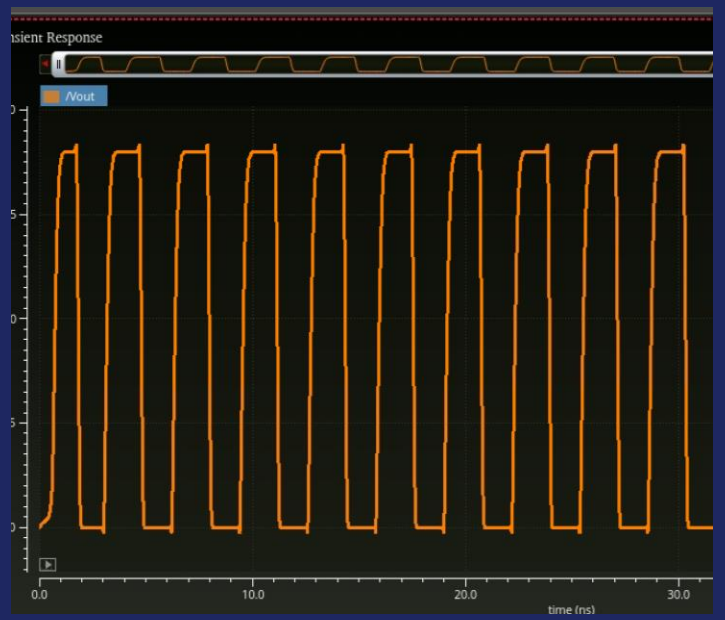
	Expression	Value TT	Value FF	Value SS
Corners	0 average(freq(clip(VT("/Vout") 30n 50n) "rising" ?xName "time" ?mode "auto" ?threshold 0.9))	490.0E6	699.9E6	313.1E6

Oscillation Frequency over Temperature

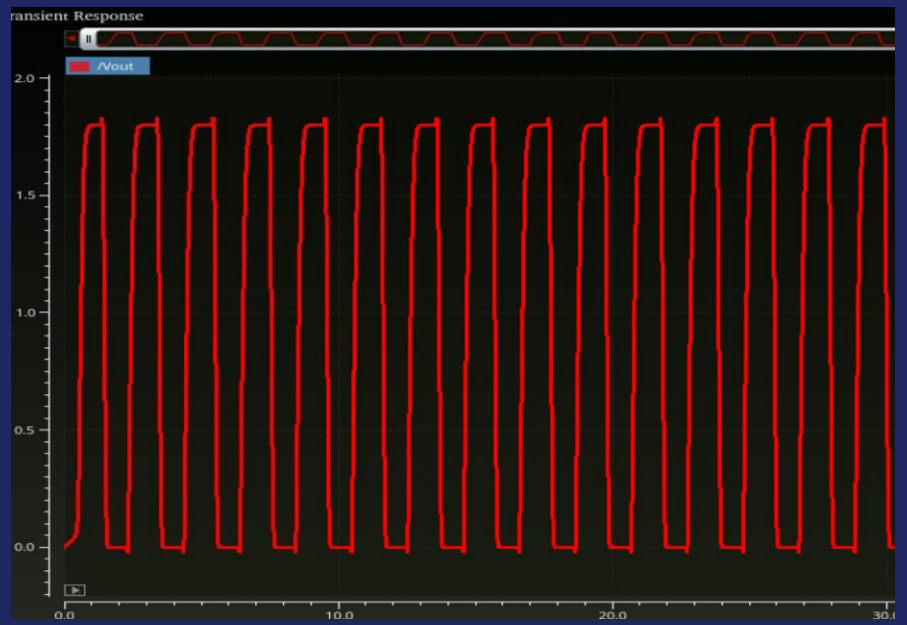
tt corner, VIN=0.327V, 0–70°C, 450–550MHz window



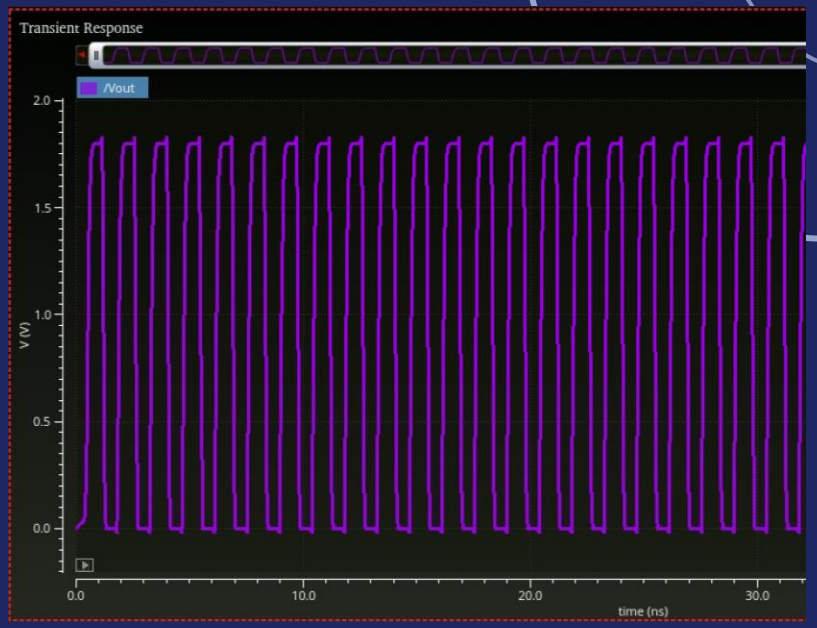
Transient Graphs and Results



SS



TT



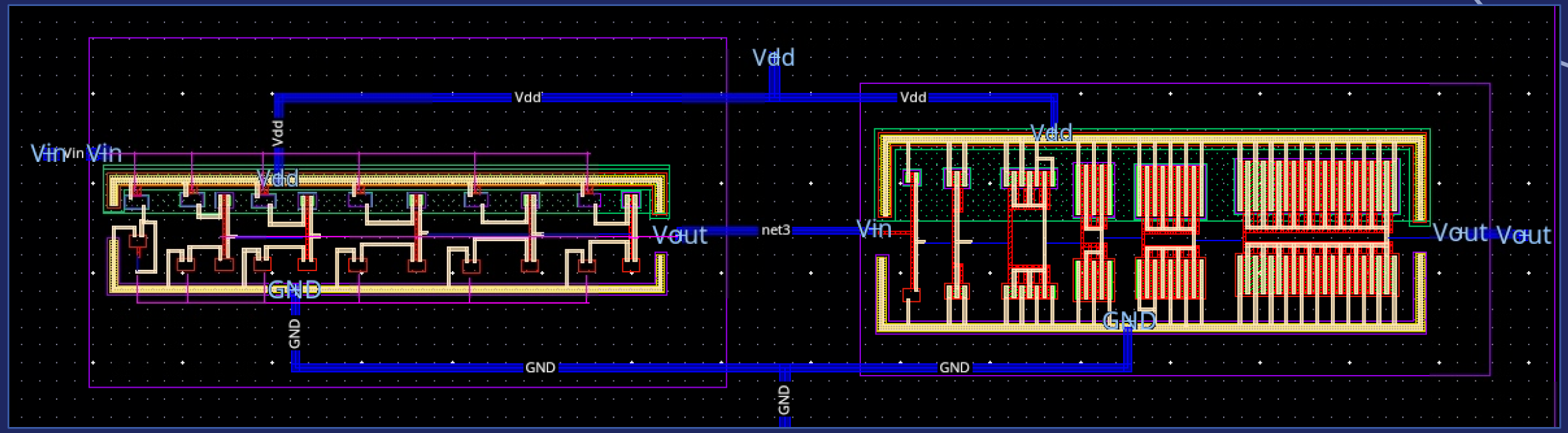
FF

Parameter	Measured	Passes?
Oscillation frequency (0–70°C, tt)	**451.2 – 549.6 MHz**	**Yes**
Oscillation frequency (ff / ss, 27°C)	**704.8 / 314.9 MHz**	**N/A**
Duty cycle (0–70°C)	**51.01 – 51.66 %**	**Yes**
Rise time, 10–90% (0–70°C)	**202.2 – 205.7 ps**	**Yes**
Fall time, 90–10% (0–70°C)	**104.9 – 125.3 ps**	**Yes**
Average power, ring only (0–70°C)	**37.6 – 47.5 µW**	**Yes**

Stage 3

Layout of the Full Design

Top cell VCO_and_Buffer: ring oscillator and output buffer, Cadence Virtuoso



VCO (left)

Five current-starved stages and the initial bias mirror, with pins Vin, Vdd, GND and Vout.

Output buffer (right)

Six-stage tapered inverter chain. Devices widen toward the output, with multiple fingers per device.

Top-level routing

Shared Vdd across the top and GND across the bottom. Net3 carries the VCO output to the buffer input.

Layout Choices and Justification

1

Guard rings

Guard rings tied to Vdd and GND avoid the need for a separate body connection.

2

Separate VCO and buffer layouts

Separate layouts allowed each DRC and LVS to be run on each circuit separately making debugging and modifications easier.

3

Top-level schematic

The top-level schematic connects both the VCO's and buffer's VDD and GND pins, accurately representing the complete system.

4

Vin and Vgs rails outside rings for VCO

Vin and the mirror node route outside the rings on met2. Vias stay inside the rings, so nothing shorts to a ring.

5

Buffer transistors with multiple fingers

Wide buffer transistors are split into fingers, keeping each stage compact as total width tapers toward the output.

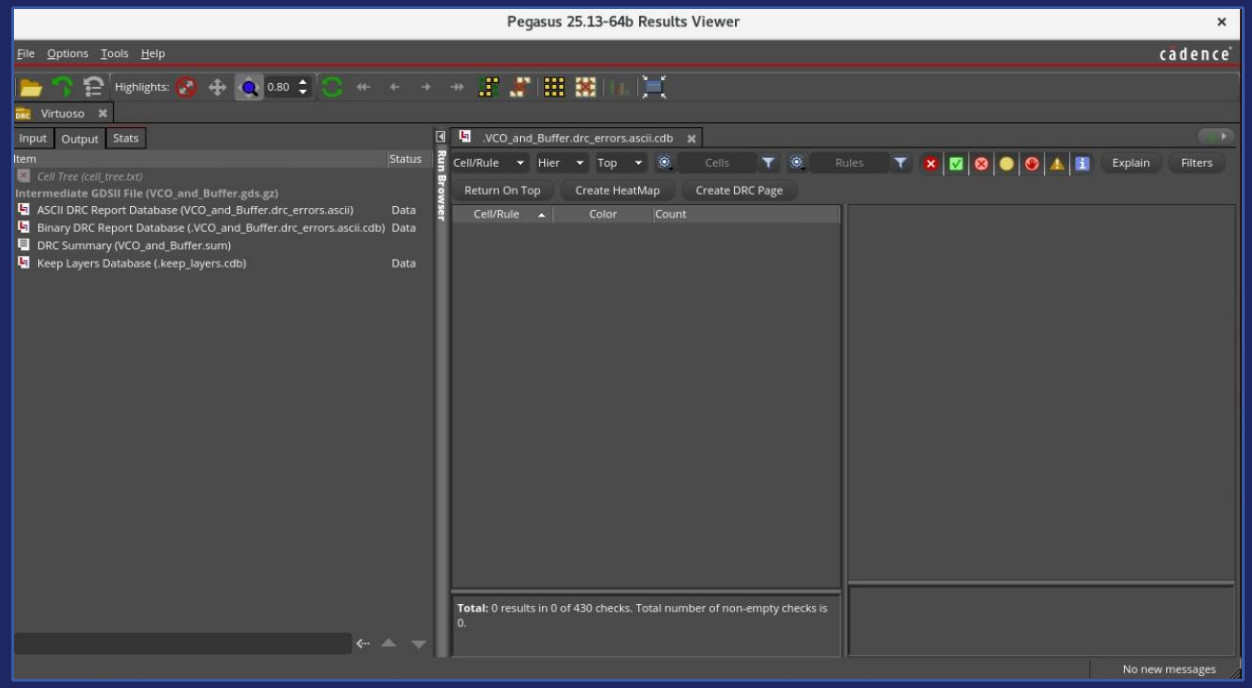
6

Net Labels

Used net labels on all pins for the VCO, buffer, and top-level layout to pass LVS.

DRC Results

Pegasus DRC with the sky130 rule deck on the top cell VCO_and_Buffer



0

DRC violations

Pegasus finished normally on the full top cell

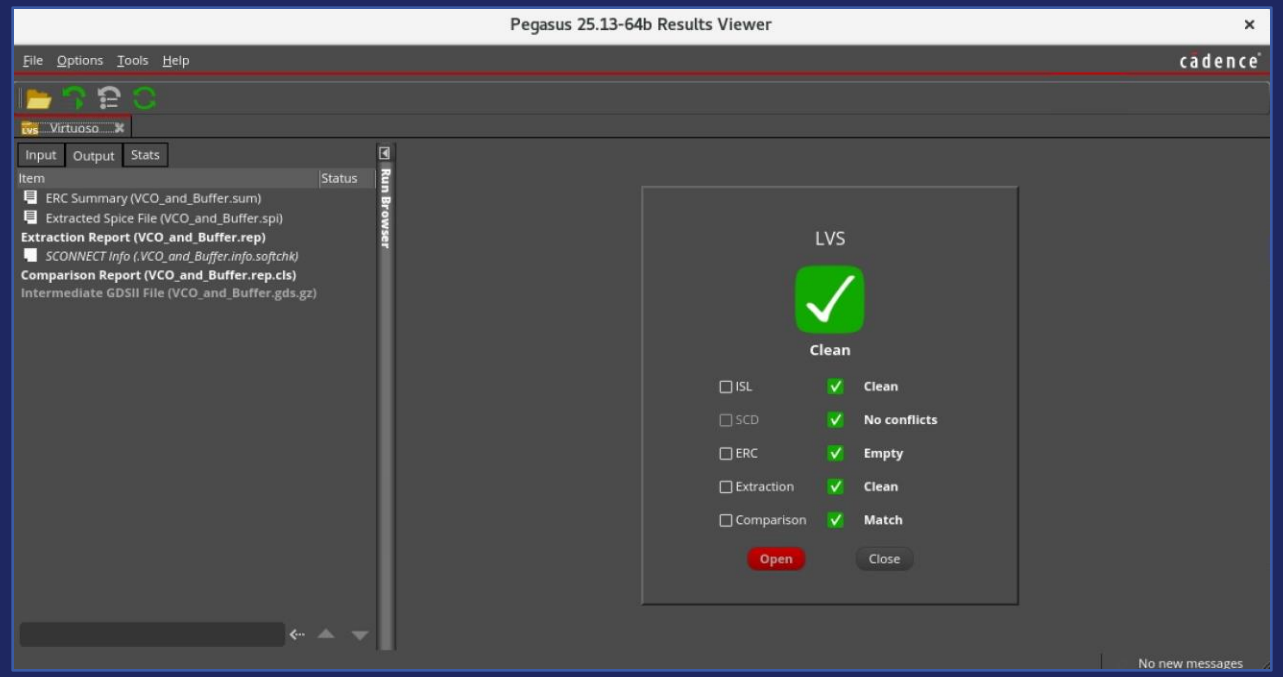
No spacing, enclosure, or tap violations were reported, so the layout passes DRC.

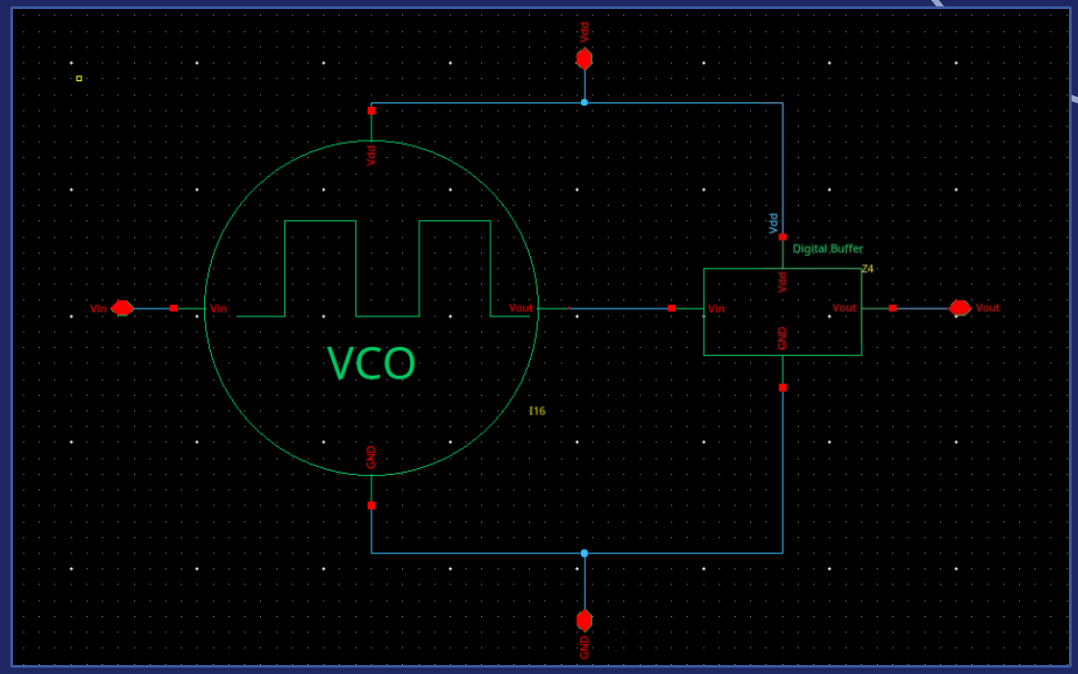
```
INFO:    All selected rules completed
         Total CPU Time                    : 3(s)
         Total Real Time                   : 10(s)
         Total Original Geometry           : 2189 (2688)
         Total DRC RuleChecks              : 431
         Total DRC Results                 : 0 (0)
         ASCII report database is /nethome/asridhar92/AMS-Onboarding/Cadence/Virtuoso/VCO_and_Buffer.drc_errors.ascii
INFO:    Checking in all SoftShare licenses.

Pegasus finished normally. 2026-10-11 18:25:30
```

LVS Results

Layout compared against the top-level schematic VCO_and_Buffer





ISL	**Clean**
SCD	**No conflicts**
ERC	**Empty**
Extraction	**Clean**
Comparison	**Match**

Top-level schematic: VCO and Digital Buffer share Vdd and GND, with pins Vin, Vdd, GND and Vout.

Final Design Review

Electrical specs from schematic simulation (tt corner, 0–70°C, VDD 1.8 V, CL 1 pF); layout verified by DRC and LVS

Specification	Requirement	Measured	Result
Oscillation frequency (0–70°C, tt)	450–550 MHz	**451.2 – 549.6 MHz**	**Pass**
Duty cycle (0–70°C)	45–55 %	**51.01 – 51.66 %**	**Pass**
Rise time, 10–90 %	≤ 250 ps	**202.2 – 205.7 ps**	**Pass**
Fall time, 90–10 %	≤ 250 ps	**104.9 – 125.3 ps**	**Pass**
Average power, ring only	< 500 µW	**37.6 – 47.5 µW**	**Pass**
DRC (Pegasus)	0 violations	**0 violations**	**Pass**
LVS (Pegasus)	Layout matches schematic	**Match**	**Pass**